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13 FEBRUARY 1980
SUPERSEDING
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8 JANUARY 1975

MILITARY SPECIFICATION

TUBE, EXTENSION, GYRO STABILIZED DRIFT METER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of extension tube for use with the MS28012 (Type B-6A) drift meter.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

WW-T-700/3E	Tube, Aluminum Alloy, Drawn, Seamless 2024
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-D-7106	Drift Meter, Electrically Driven Gyro Stabilized
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series. General Specificaton For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6605

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence For The Selection Of
MIL-STD-794	Parts and Equipment, Procedures For Packaging and Packing Of
MS28012	Drift Meter, Electrically Driven, Gryo Stabilized

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN5767	Tube-Extension, Gyro Stabilized Drift Meter
AN5768	Rod Assembly-Extension, Drift Meter Prism Control

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.1.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the extension tube, except where magnetic materials are essential.

3.1.3 Protective treatment. When materials are used in the construction of the tube that are subject to corrosion in salt air or other atmospheric conditions likely to occur during usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions, shall be avoided. Cadmium plating shall not be used.

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3.1.4 Selection of material. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

* 3.1.4.1 Standard parts. Standard parts (AN or MS) shall be used. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or MS Standard parts without alteration, and provided the corresponding AN or MS part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is not suitable corresponding AN or MS Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

* 3.2 Design. The extension tube shall conform to Drawing AN5767, P/N AN5767-2.

3.3 Construction. The extension tube shall be constructed to meet the requirements specified in Specificaton MIL-D-7106 where applicable, and shall be fabricated of aluminum-alloy tubing conforming to Specification WW-T-700/3E, type I, condition T.

* 3.3.1 Upper mount. The upper end of extension tube shall be constructed to permit mounting to the lower end of the rotating tube of the MS28012 drift meter.

* 3.3.2 Lower mount. The lower end of the extension tube shall be constructed to permit mounting to the upper end of the outer sleeve of the MS28012 drift meter.

3.3.3 The extension tube shall increase the over-all length of the drift meter by 5 feet.

* 3.3.4 Prism control-rod. A prism control-rod assembly in accordance with Drawing AN5768, P/N AN5768-6 shall be provided.

3.3.5 Guides. Guides shall be provided in the extension tube, for support or guidance of the rods connecting to cables operating the index prism.

* 3.3.6 Optical abberations or misalignments. It shall be possible to attach the extension tube to the MS28012 drift meter with a tube length of at least 30 inches without introducing optical aberrations or misalignments which would affect the accuracy and performance of the drift meter.

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* 3.3.6.1 Attaching holes. The upper and lower ends of the extension tube shall be supplied with the necessary holes for attaching the MS28012 drift meter without requiring a physical change, i.e., drilling holes, tapping of threads, or rework of any kind, of the extension tube or drift meter.

3.3.6.2 Lenses. The extension tube shall be supplied with necessary lenses to provide drift meters with tube lengths from 90 to 153 inches with a field of view not less than 25 degrees at unity power.

3.3.7 Resilient construction. The tube shall be so constructed that no parts will work loose in service and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.3.8 Collimation. Addition of the extension tube shall not alter collimation of the drift meter, i.e., in a drift meter which was correctly collimated before adding the extension tube, the reticle and axial target images shall remain free of paralax after adding the extension tube.

3.3.9 Power. Addition of the extension tube shall not alter the power of the drift meter more than 5 percent.

3.3.10 Exit pupil. Addition of the extension tube shall not decrease the exit pupil below 4 mm diameter.

3.3.11 Curvature of field. Addition of the extension tube shall not increase the curvature of field at 10 degrees from the center of the apparent field at low power by more than 1/2 diopter.

3.3.12 Astigmatism. Addition of the extension tube shall not increase the astigmatism at 10 degrees from the center of the apparent field at low power by more than 1/2 diopter.

3.3.13 Distortion. Addition of the extension tube shall result in the minimum practicable increase in distortion.

3.4 Interchangeability. All parts have the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-STD-100.

3.5 Finish. All surfaces of the extension tube shall have a durable, dull black finish.

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* 3.5.1 Aluminum finish. Aluminum-alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except that the inside surface surrounding small holes need not be anodized.

3.6 Screw-threads. Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.7 Weight. The weight of the extension tube completely assembled shall not exceed 8 pounds.

3.8 Identification of product.

3.8.1 Nameplate. The nameplate shall be securely attached to the exterior of the tube and shall be legibly and durably marked with the exterior of the tube and shall be legibly and durably marked with the information listed below. The information marked in the spaces provided on the nameplate shall be in accordance with Standard MIL-STD-130.

TUBE, EXTENSION, GYRO STABILIZED DRIFT METER
AN5767- (Proper Dash Number)
Stock No.
Manufacturer's Part No.
Contract or Order No.
Manufacturer's Serial No.
Manufacturer's Name or Trade-Mark
U S Property

3.9 Mounting screws. The contractor shall furnish sufficient mounting screws for attaching the extension tube to the drift meter.

3.9.1 Envelope. An envelope, containing the mounting screws, shall be furnished with each extension tube and shall be attached to the tube by means of a wire wrapped around the tube. The following information shall be printed on the face of the envelope.

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.10 Workmanship. The extension tube including all parts and accessories shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to insure that the lenses are free from scratches and dirt and properly assembled. There shall be no burrs or sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. All the tests required for the testing of the tubes are classified as Inspection Tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Individual tests. Each tube shall be subjected to the following tests:

4.2.1 Examination of product. Each tube shall be carefully examined to determine conformance with this specification and with the dimensions and tolerances shown on the drawings.

4.2.2 Functional. Two each drift meters with 40-inch tube lengths shall be inspected to determine conformance with Specification MIL-D-7106. Drift meters not conforming to this specification shall be made to do so. One of the instruments shall be disassembled at the junction of sleeve and rotating tube. Both of the instruments shall be placed in appropriate test fixtures and provided with collimators at the objective end of instruments. Each extension tube shall be inserted between the two parts of disassembled drift meter. The collimator images see through the two drift meters shall be compared visually. The quality of the two images shall compare favorable with due allowances being given for the increase aberrations and loss in the light transmissions accounted by the additonal lenses in the extension tube. No change in the amount of parallax between the axial images of collimator and reticle shall occur. If the image is not satisfactory, or additional parallax results, the extension tube shall be rejected.

4.2.3 Real field. Using the unit power eyepiece, the field of view of each drift meter shall be measured with the extension tube inserted. If field of view of the instrument is less than 25 degrees, the extension tube shall be rejected.

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4.3 Rejection and retest. Any extension tube failing to meet the requirements of this specification shall be rejected. Tubes which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects, shall be furnished the Inspector. Tubes rejected after retest shall not be resubmitted without the specific approval of the Inspector or the procuring agency.

* 4.4 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

* 5.1 Preservation and packaging.

* 5.1.1 Level A. (Reference MIL-STD-794 for definition of levels of preservation and packaging.) Tube, Extension, Gyro Stabilized Drift Meter shall be packaged in quantity unit packs of one each in accordance with the appropriate method of preservation (See Appendix D of MIL-STD-794) and accomplished as specified in MIL-P-116. Appropriate unit container to be selected from Table I (Exterior container) MIL-STD-794. Sufficient cushioning shall be applied to provide the protection necessary for completed pack to meet the rough handlng drop test requirements of MIL-P-116. These preservation and packaging requirements apply unless otherwise specified by the procuring agency.

* 5.1.2 Level C. Same requirements as Level A above except Method III of MIL-P-116 applies in lieu of method selected from the appropriate table of MIL-STD-794.

* 5.2 Packing.

* 5.2.1 Level A. (Reference MIL-STD-794 for definition of levels of packing) Tube, Extension, Gyro Stabilized Drift Meter packaged as specified in 5.1.1 shall be packed in weather-resistant shipping container (See Table I, MIL-STD-794) unless otherwise specified by the procuring agency. Insofar as practical, exterior shipping containers shall be of uniform shape, size, of minimum tar and cube consistant with the protection required.

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* 5.2.2 Level B. All units packaged as specified in 5.1.1 shall be packed in domestic type exterior shipping containers (See Table I, MIL-STD-794). Otherwise requirements as specified in 5.2.1 shall apply unless otherwise specified by the procuring agency.

* 5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

* 5.3 Marking and labeling. All markings shall be in accordance with the requirements of Standard MIL-STD-129 and as specified herein.

5.3.1 Packages. Each unit package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the package is opened.

TUBE, EXTENSION, GYRO STABILIZED DRIFT METER
AN5767- (Proper Dash Number)
Stock No.
Manufacturer's Part No.
Contract or Order No.
Quantity
Name of Manufacturer
Name of Contractor (if different from manufacturer)

5.3.2 Precautionary markings. The following precautionary marking shall appear on two opposite sides of each interior package, whenever practicable, depending on the size of the carton, and on the shipping container.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

* 6.1 Intended use. The extension tube covered by this specification are intended for use with the MS28012 gyro stabilized drift meter when a drift meter with the tube length greater than 100 inches is required.

6.2 Reclaimed material. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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* 6.3 Margin indication. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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